

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

PRELAB EXERCISES (3 points)

This prelab must be completed and handed in to the lab TA at the start of the lab.

Question 1

1 point

Why is it important to recharge the spheres before each measurement?

Question 2

1 point

Compute the force between one gram of protons and one gram of electrons that are 10 km apart. Show your work and explain your steps.

Question 3**1 point**

Consider the function $y = x^n$ when $n = -2$. Fill out the table on the left for the given values of x . Plot $\ln x$ vs. $\ln y$ on the right. Make sure you draw your coordinate axes in a smart way. Explain how to obtain n from the graph. (Hint: can you fit a straight line to your plotted points?)

x	y	$\ln x$	$\ln y$
1			
2			
3			
4			
5			
6			

